



METABOLIC & GLP-1

Liraglutide

Liraglutide – 5 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Liraglutide
SYNONYMS / ALIASES	NN2211; GLP-1 receptor agonist analog
CAS NUMBER	204656-20-2
MOLECULAR FORMULA	C172H265N43O51
MOLECULAR WEIGHT	3751.2 g/mol
AMINO ACID SEQUENCE	HAEGTFTSDVSSYLEGQAAKEFIAWLVRGRG (GLP-1(7-37) backbone with Arg at position 34; Lys at position 26 conjugated to a C16 palmitoyl chain via a gamma-Glu spacer)
SOURCE	Synthetic (solid-phase peptide synthesis with side-chain acylation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	5 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. Palmitoylated peptides self-associate into heptamers and are sensitive to interface aggregation - handle with minimal agitation.

STRUCTURAL NOTES

Liraglutide is a synthetic analog of human glucagon-like peptide-1 based on the GLP-1(7-37) sequence, sharing approximately 97% backbone homology with native GLP-1. It carries an arginine substitution at position 34 and a C16 palmitoyl fatty acid attached to the epsilon-amino group of Lys26 via a gamma-glutamyl spacer. The acyl modification promotes reversible albumin binding and self-association, extending its circulating half-life relative to native GLP-1.

SELECTED PEER-REVIEWED REFERENCES

1. Knudsen L.B. et al. - *Potent derivatives of glucagon-like peptide-1 with pharmacokinetic properties suitable for once daily administration* - Journal of Medicinal Chemistry, 2000 - DOI: 10.1021/jm9909645
2. Marso S.P. et al. - *Liraglutide and cardiovascular outcomes in type 2 diabetes* - New England Journal of Medicine, 2016 - DOI: 10.1056/NEJMoa1603827
3. Pi-Sunyer X. et al. - *A randomized, controlled trial of 3.0 mg of liraglutide in weight management* - New England Journal of Medicine, 2015 - DOI: 10.1056/NEJMoa1411892

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.